Computer System Overview

Chapter 1

Basic Elements of Computer

- Processor(处理器，CPU-中央处理单元)
- Main Memory(内存/主存)
  - Volatile(易失的)，referred to as real memory or primary memory or physical main memory
- I/O modules(I/O模块)
  - secondary storage devices(辅助存储设备)
  - communications equipment
  - Terminals(终端)
- System bus(系统总线)
  - communication among processors, memory, and I/O modules

Computer Components: Top-Level View

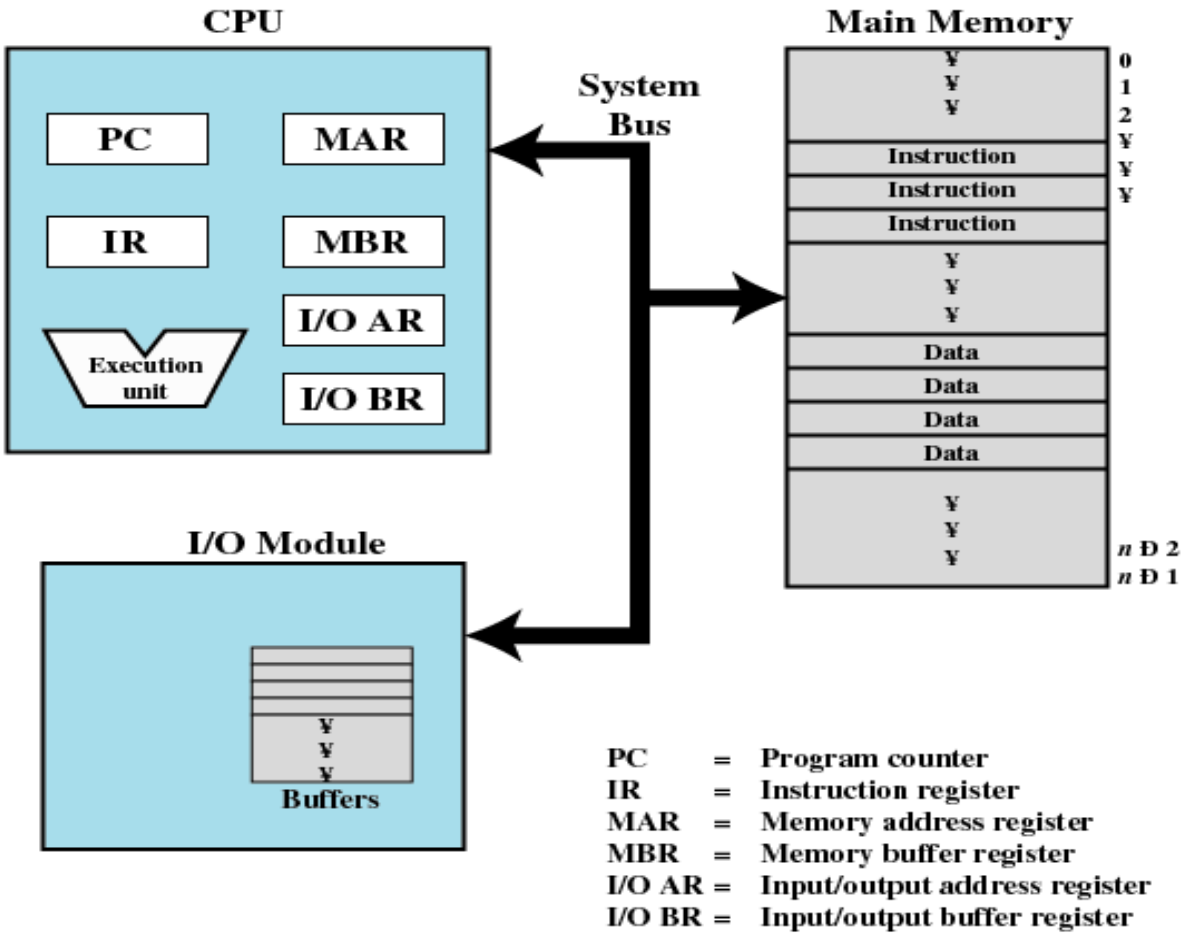
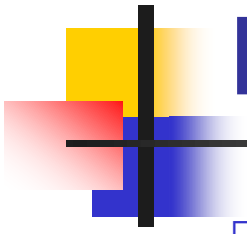


Figure 1.1 Computer Components: Top-Level View

Processor Registers(寄存器)

- Memory address register (MAR)
  - Specifies the address for the next read or write
- Memory buffer register (MBR)
  - Contains data written into memory or receives data read from memory
- I/O address register
- I/O buffer register



Processor Registers

- User-visible registers
  - Enable programmer to minimize main-memory references by optimizing register use
- Control and status registers
  - Used by processor to control operating of the processor
  - Used by privileged operating-system routines to control the execution of programs

User-Visible Registers (1)

- May be referenced by machine language
- Available to all programs - application programs and system programs
- Types of registers
  - Data
  - Address
    - Index
    - Segment pointer
    - Stack pointer

User-Visible Registers (2)

- Address Registers
  - Index
    - Involves adding an index to a base value to get an address
  - Segment pointer
    - When memory is divided into segments, memory is referenced by a segment and an offset
  - Stack pointer
    - Points to top of stack

Control and Status Registers (1)

- Program Counter (PC, 程序计数器)
  - Contains the address of an instruction to be fetched
- Instruction Register (IR)
  - Contains the instruction most recently fetched
- Program Status Word (PSW, 程序状态字)
  - Condition codes
  - Interrupt enable/disable
  - Supervisor/user mode

Control and Status Registers (2)

- Condition Codes or Flags
  - Bits set by the processor hardware as a result of operations
  - Example: arithmetic operation
    - Positive result
    - Negative result
    - Zero
    - Overflow

Instruction Execution(指令执行)

- Two steps
  - Processor reads instructions from memory
  - Processor executes each instruction
- Instruction cycle(指令周期)

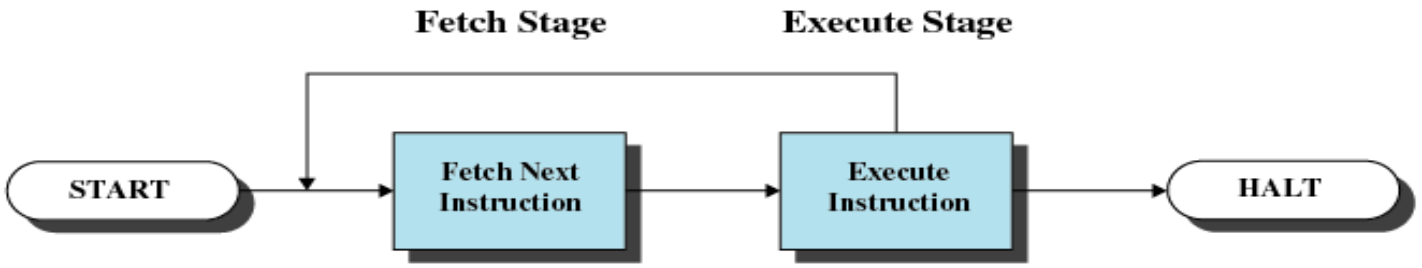


Figure 1.2 Basic Instruction Cycle

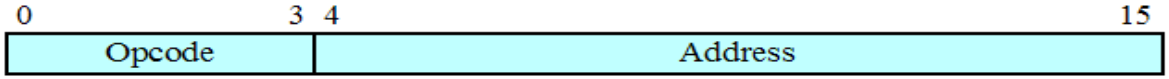
Instruction Fetch and Execution

- Program counter (PC, 程序计数器) holds address of the instruction to be fetched next
- The processor fetches the instruction (取指令) from memory
- Program counter is incremented after each fetch

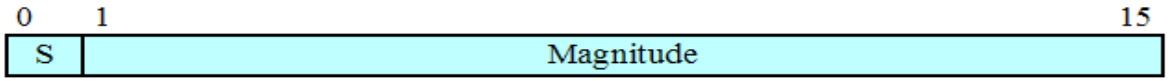
Instruction Register

- Fetched instruction is placed in the instruction register (指令寄存器)
- Categories of actions:
  - Processor-memory (处理器和内存之间)
    - Transfer data between processor and memory
  - Processor-I/O (处理器和I/O设备之间)
    - Data transferred to or from a peripheral device(外部设备)
  - Data processing
    - Arithmetic(算术) or logic operation on data
  - Control
    - Alter sequence (顺序) of execution

Characteristics of a Hypothetical Machine



(a) Instruction format



(b) Integer format

Program Counter (PC) = Address of instruction
Instruction Register (IR) = Instruction being executed
Accumulator (AC) = Temporary storage

(c) Internal CPU registers

0001 = Load AC from Memory
0010 = Store AC to Memory
0101 = Add to AC from Memory

(d) Partial list of opcodes

Figure 1.3 Characteristics of a Hypothetical Machine

Example of Program Execution

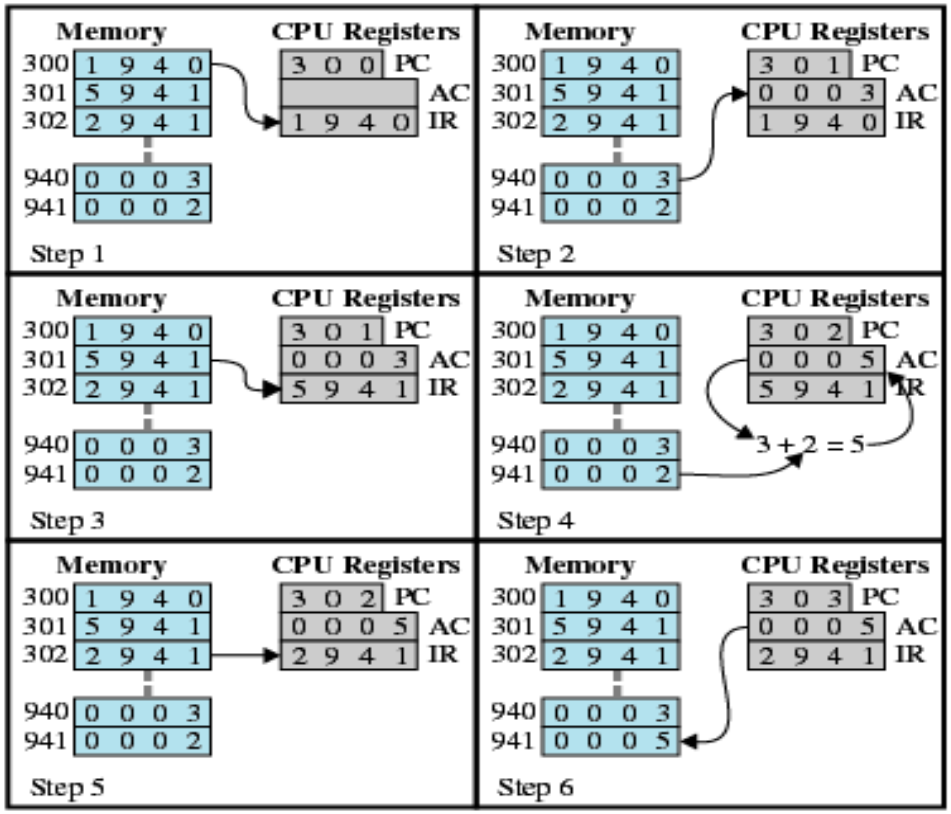


Figure 1.4 Example of Program Execution
(contents of memory and registers in hexadecimal)

Interrupts (中断)

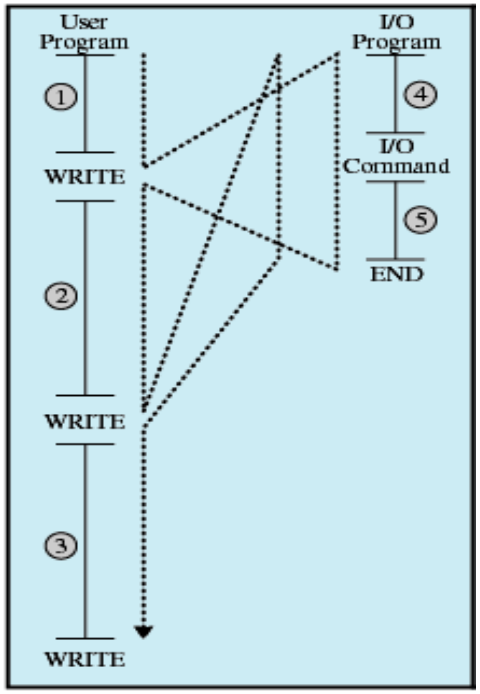
- Most I/O devices are slower than the processor
  - Processor must pause to wait for device
- Interrupts are provided primarily as a way to improve processor utilization
- Interrupt the normal sequencing of the processor

Classes of Interrupts

Table 1.1 Classes of Interrupts

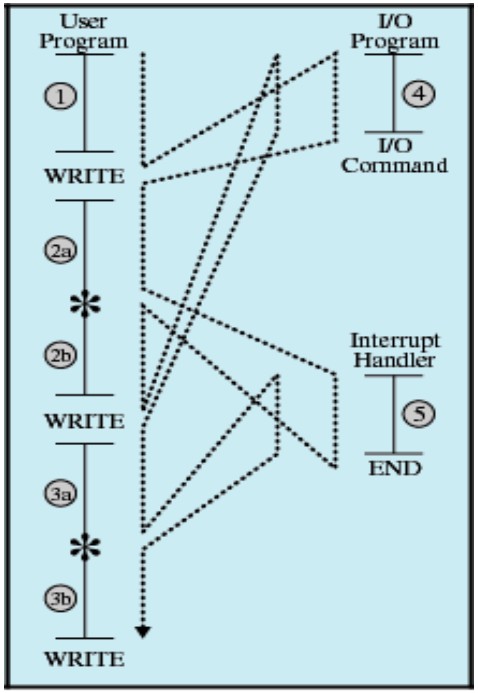
Program	Generated by some condition that occurs as a result of an instruction execution, such as arithmetic overflow, division by zero, attempt to execute an illegal machine instruction, and reference outside a user's allowed memory space.
Timer	Generated by a timer within the processor. This allows the operating system to perform certain functions on a regular basis.
I/O	Generated by an I/O controller, to signal normal completion of an operation or to signal a variety of error conditions.
Hardware failure	Generated by a failure, such as power failure or memory parity error.

Program Flow of Control Without Interrupts



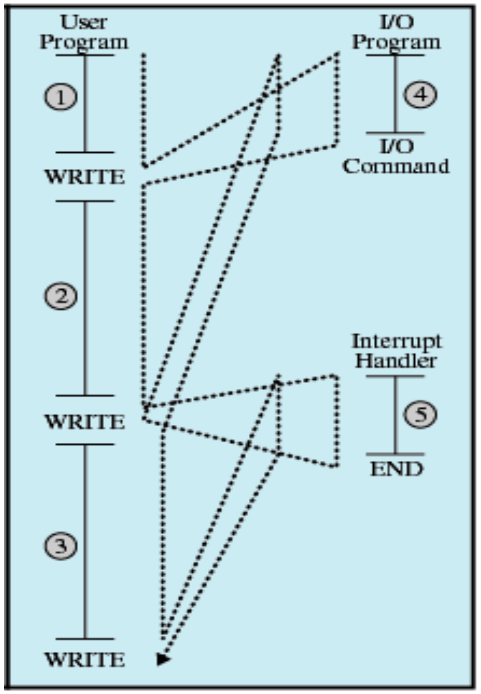
(a) No interrupts

Program Flow of Control With Interrupts, Short I/O Wait



(b) Interrupts; short I/O wait

Program Flow of Control With Interrupts, Long I/O Wait



(c) Interrupts; long I/O wait

Interrupt Handler(中断服务程序)

- Program to service a particular I/O device
- Generally part of the operating system
- Interrupt Service Routine (ISR, 中断服务程序)

Interrupts

- Suspends (挂起) the normal sequence of execution

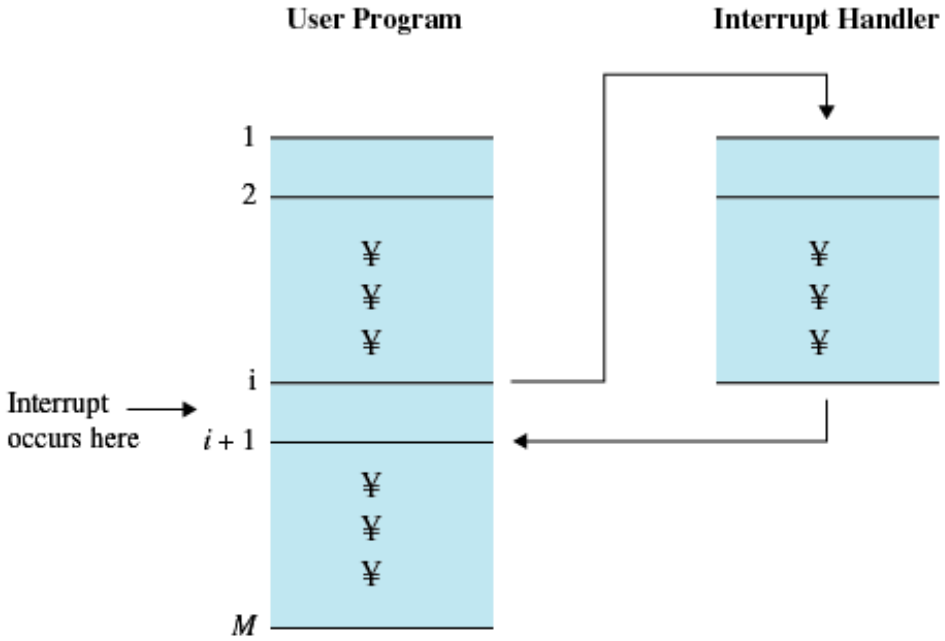


Figure 1.6 Transfer of Control via Interrupts

Interrupt Cycle

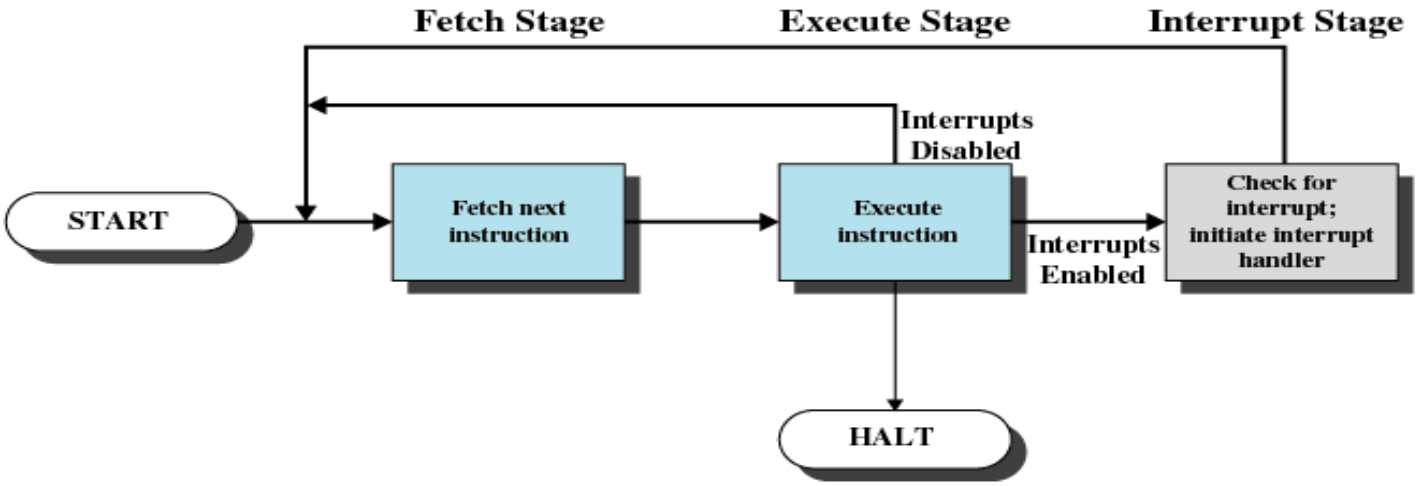
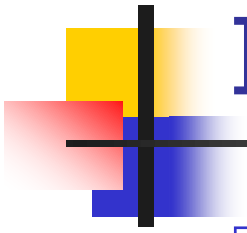


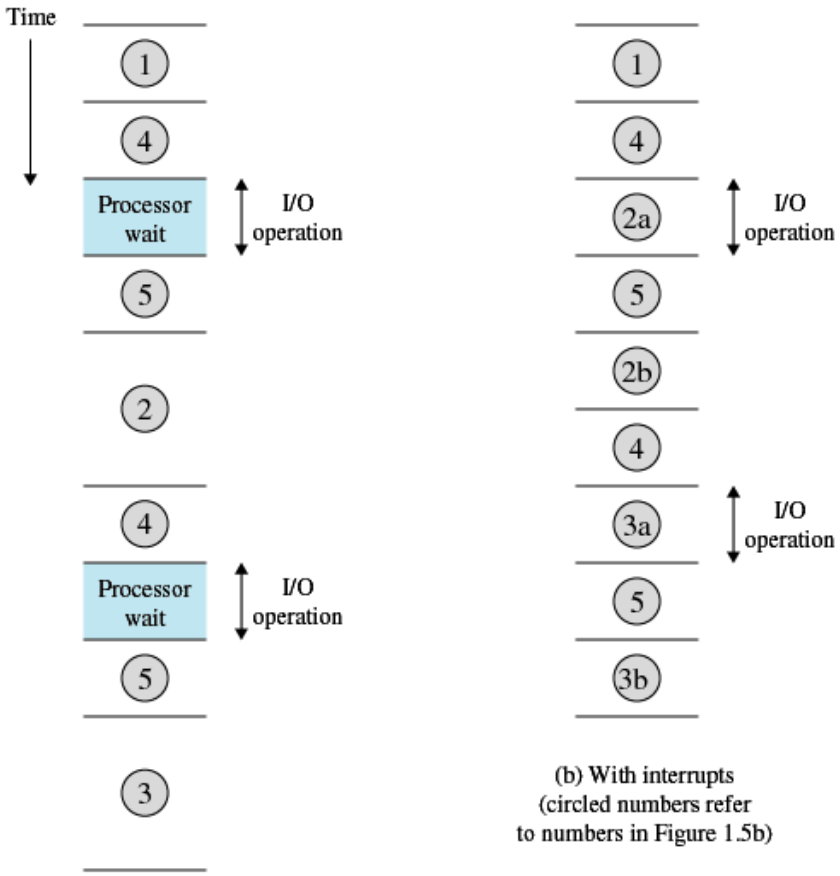
Figure 1.7 Instruction Cycle with Interrupts



Interrupt Cycle

- Processor checks for interrupts after each instruction
- If no interrupts are pending, fetch the next instruction for the current program
- If an interrupt is pending, suspend execution of the current program, and execute the interrupt-handler routine

Timing Diagram Based on Short I/O Wait

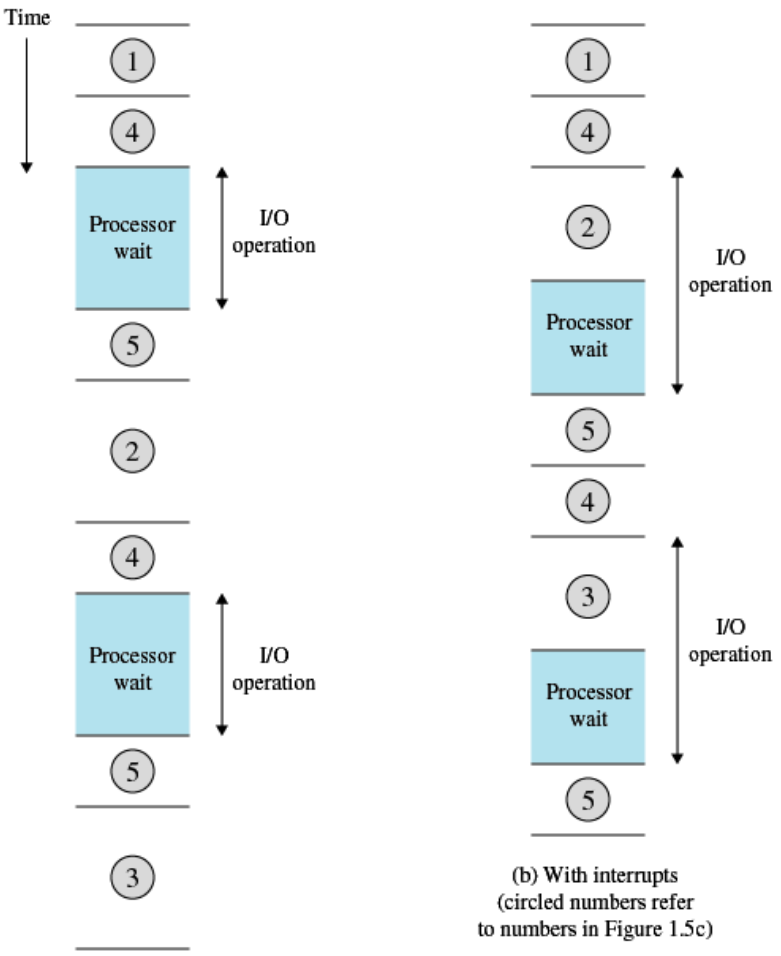


(a) Without interrupts (circled numbers refer to numbers in Figure 1.5a)

(b) With interrupts (circled numbers refer to numbers in Figure 1.5b)

Figure 1.8 Program Timing: Short I/O Wait

Timing Diagram Based on Short I/O Wait



(a) Without interrupts (circled numbers refer to numbers in Figure 1.5a)

(b) With interrupts (circled numbers refer to numbers in Figure 1.5c)

Figure 1.9 Program Timing: Long I/O Wait

Simple Interrupt Processing

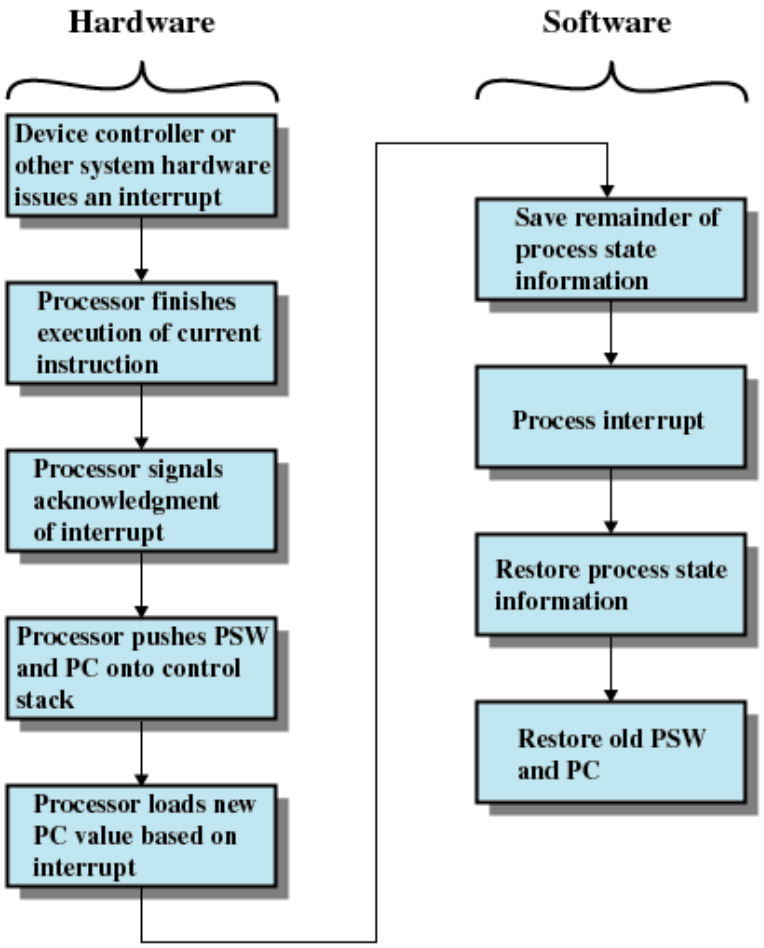
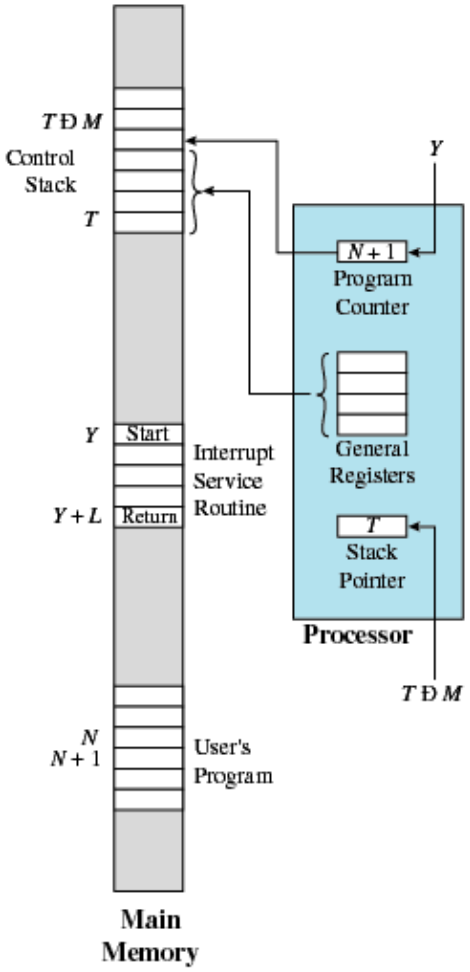


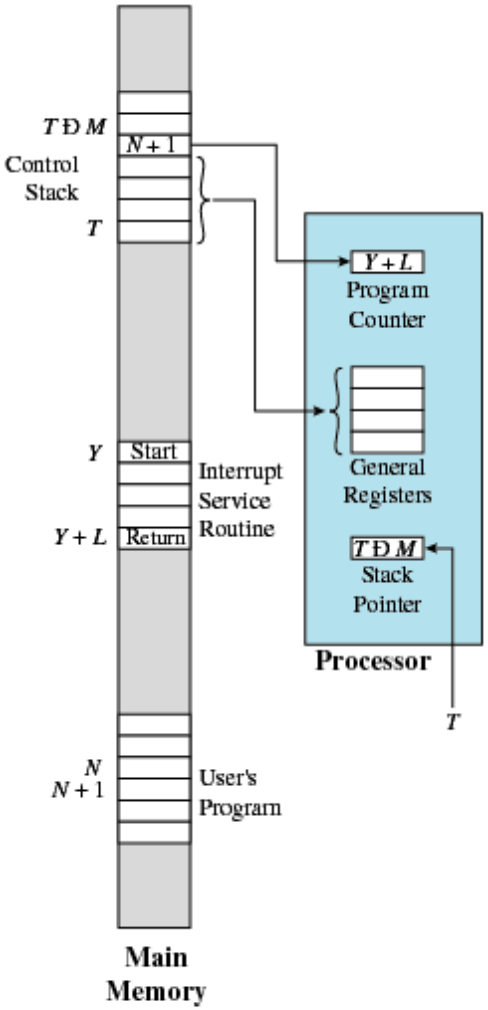
Figure 1.10 Simple Interrupt Processing

Changes in Memory and Registers for an Interrupt



(a) Interrupt occurs after instruction at location N

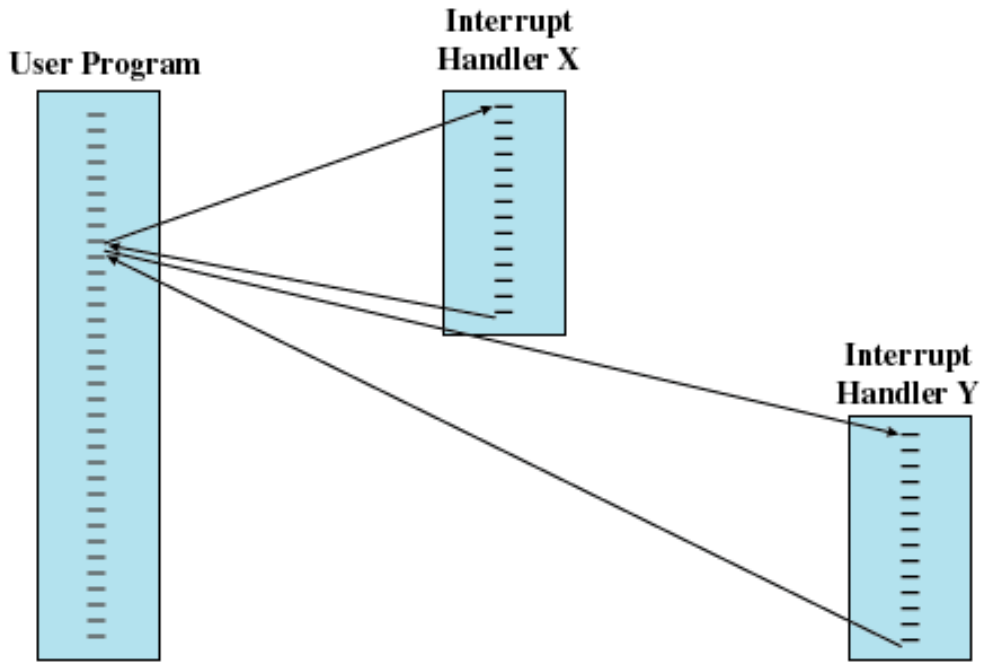
Changes in Memory and Registers for an Interrupt



(b) Return from interrupt

Multiple Interrupts

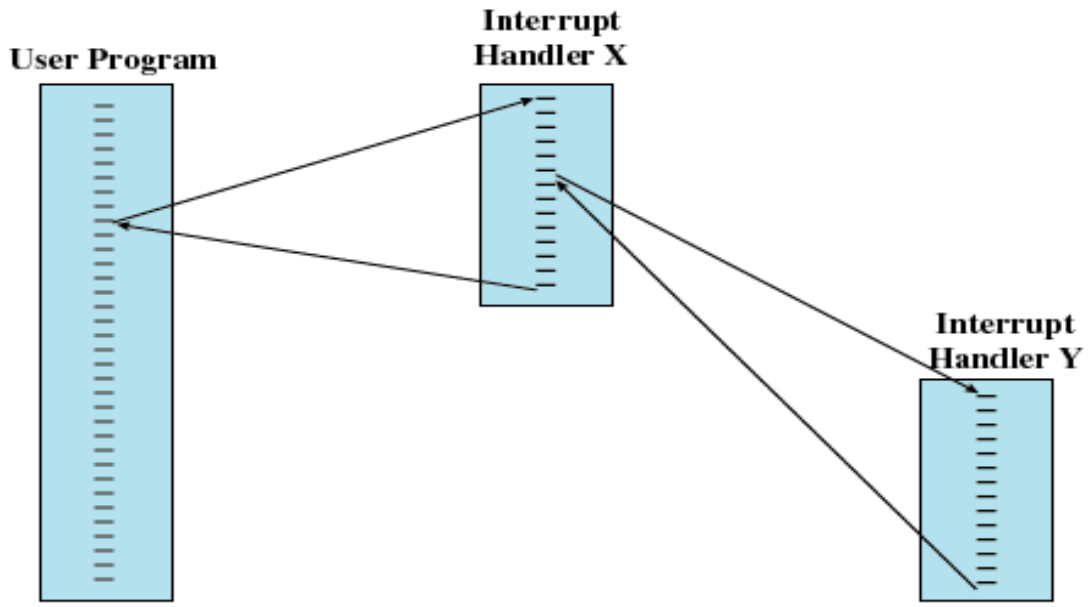
- Disable interrupts while an interrupt is being processed



(a) Sequential interrupt processing

Multiple Interrupts

- Define priorities (优先级) for interrupts



(b) Nested interrupt processing

Multiple Interrupts

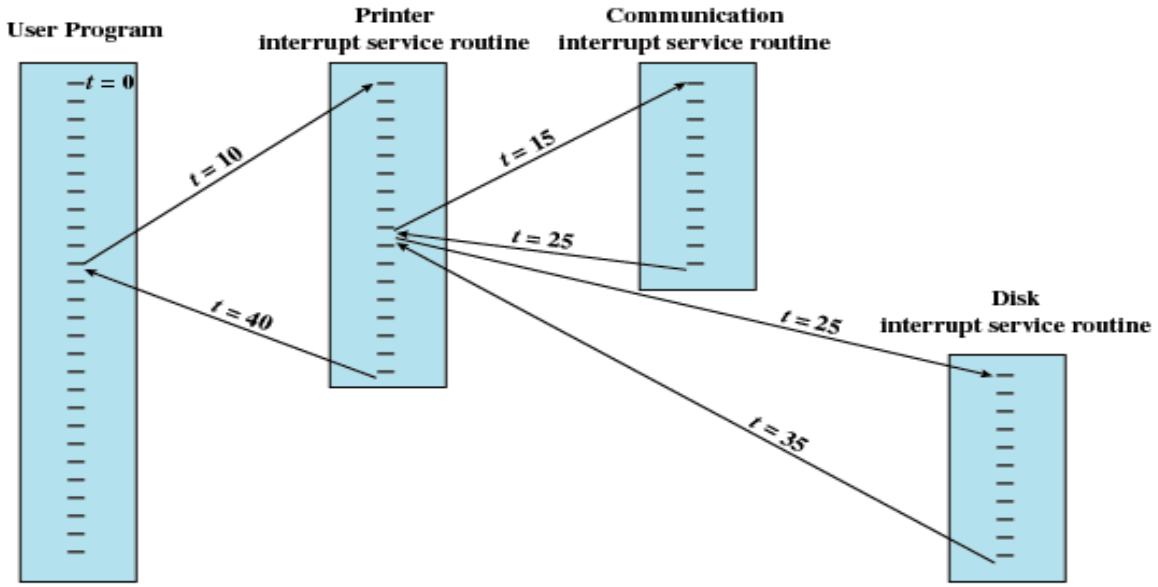
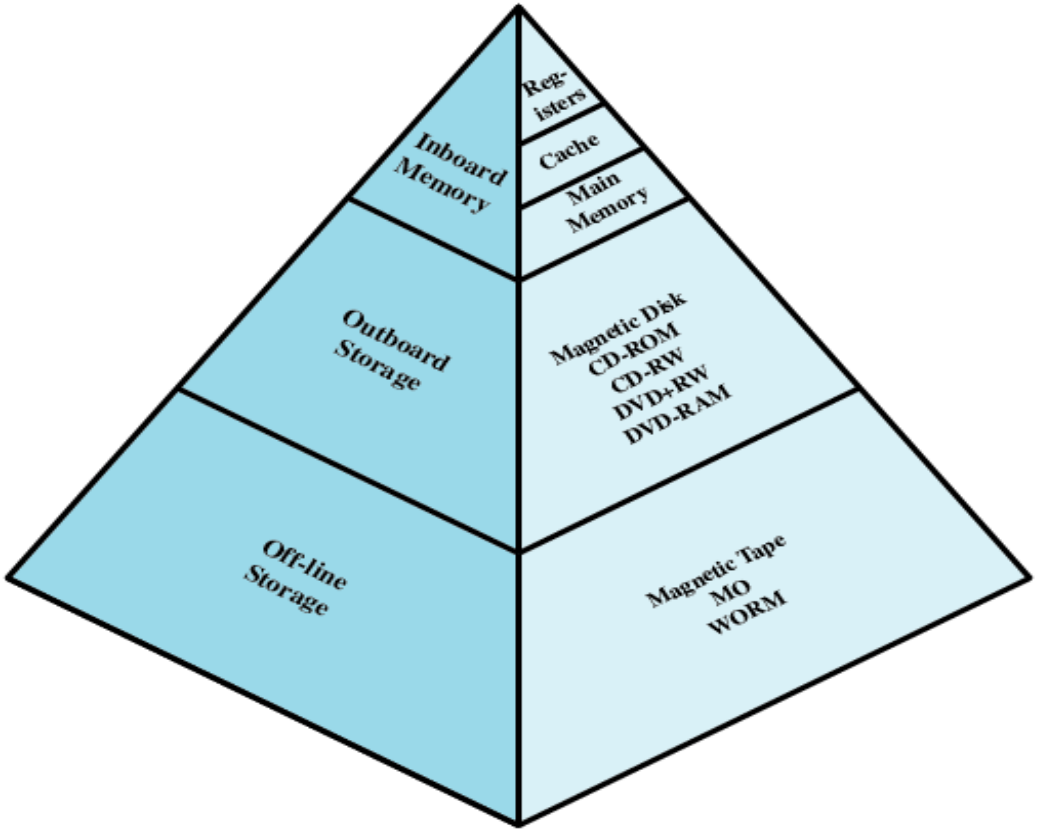


Figure 1.13 Example Time Sequence of Multiple Interrupts

Memory Hierarchy(存储层次)

- Three key characteristics of memory
  - Capacity, access time and cost
- Faster access time, greater cost per bit
- Greater capacity, smaller cost per bit
- Greater capacity, slower access speed

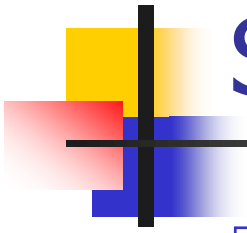
Memory Hierarchy



Going Down the Hierarchy

- Decreasing cost per bit
- Increasing capacity
- Increasing access time
- Decreasing frequency of access of the memory by the processor

Secondary Memory (辅助存储器)



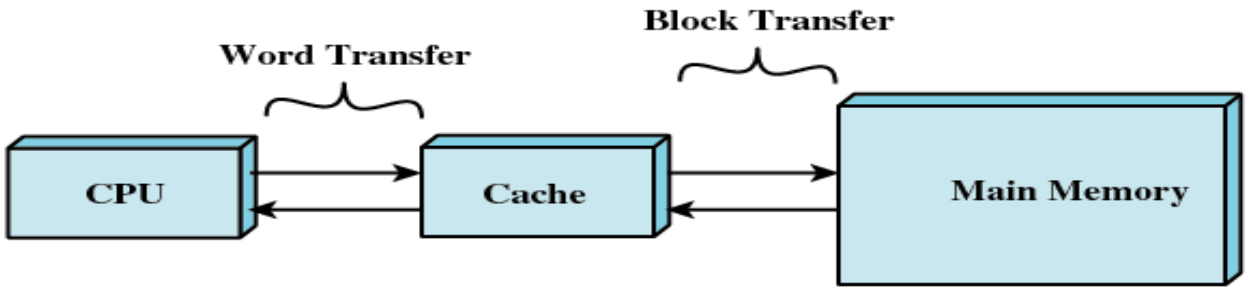
- Used to store program and data
- Nonvolatile (非易失的)
- Auxiliary memory (辅助存储)

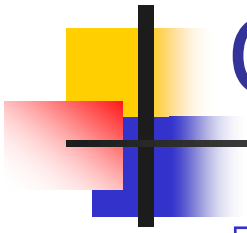
Cache Memory(高速缓存)

- On all instruction cycle, the processor accesses main memory at least once, to fetch the instruction
- Persistent mismatch between processor and main memory speed
  - Processor speed is faster than memory speed
- A small, fast memory between the processor and main memory is provided

Cache Memory(高速缓存)

- Exploit the principle of locality
- Cache memory is invisible to operating system and programmer





Cache Memory

- Contains a copy of a portion of main memory
- Processor first checks cache
- If not found in cache, the block of memory containing the needed information is moved to the cache and delivered to the processor

Cache/Main Memory System

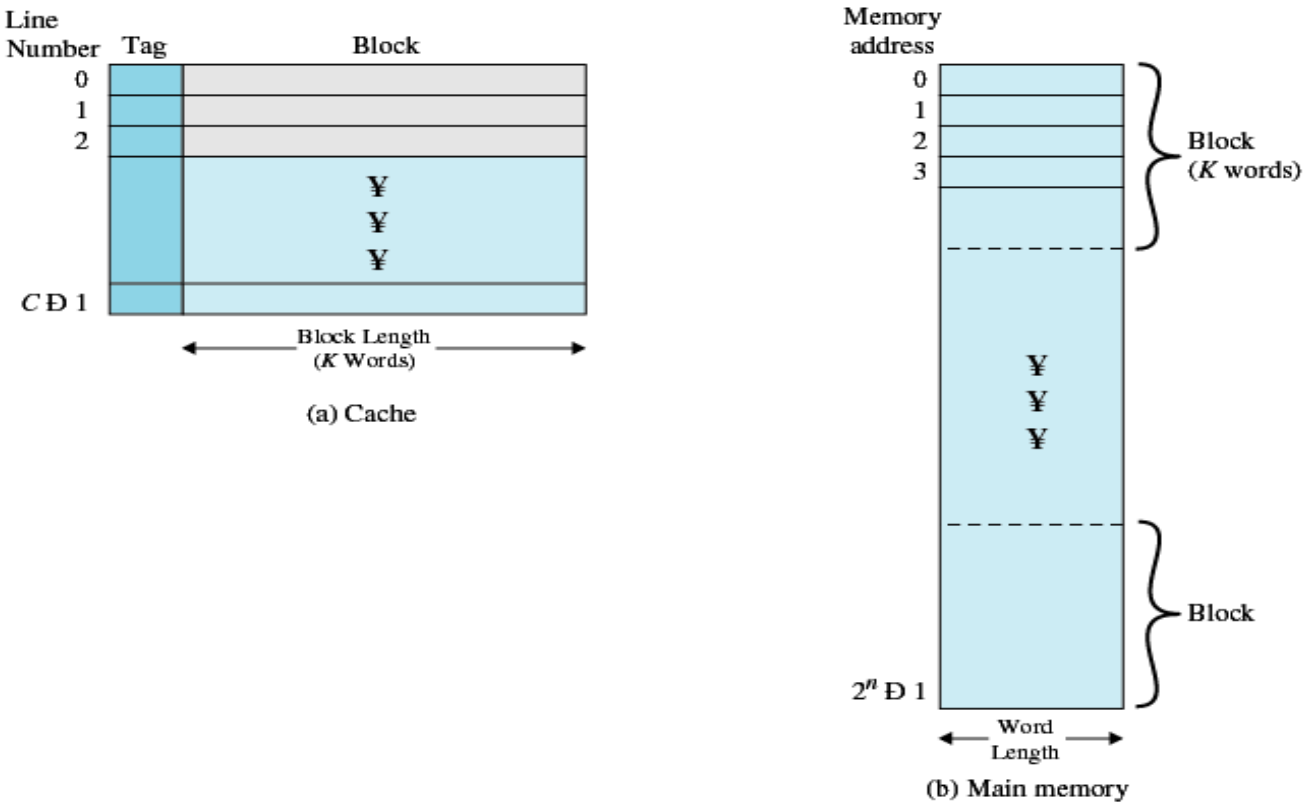


Figure 1.17 Cache/Main-Memory Structure

Cache Read Operation

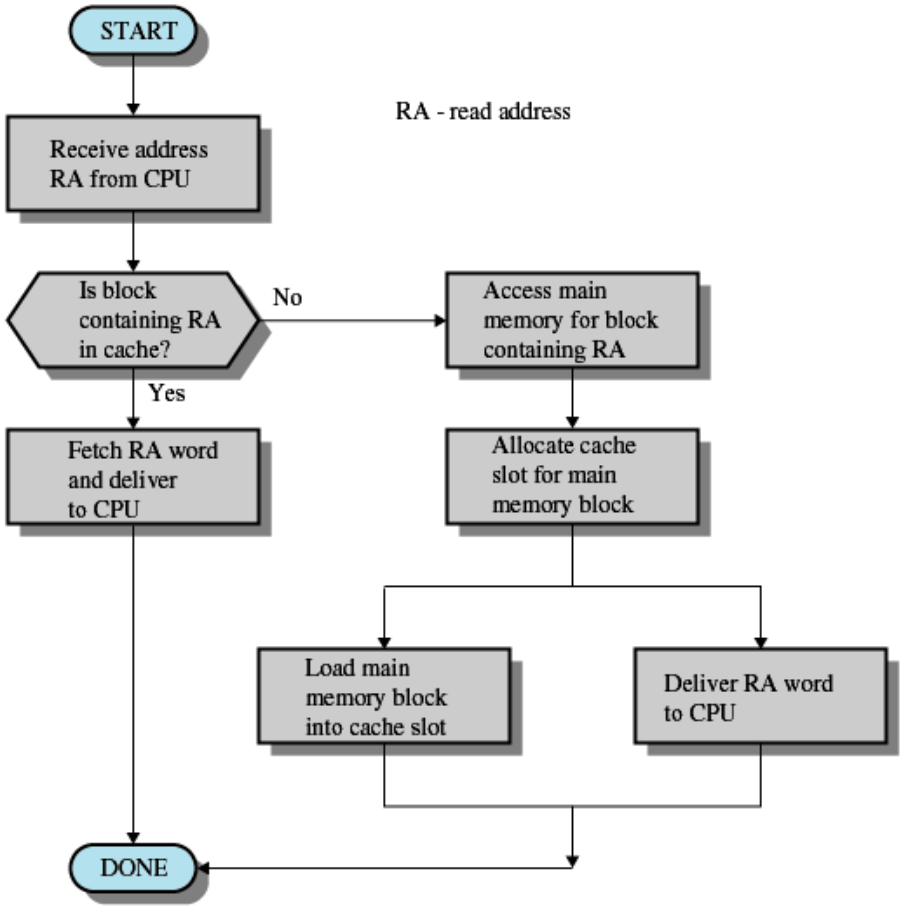
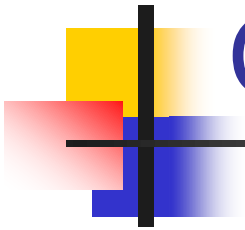
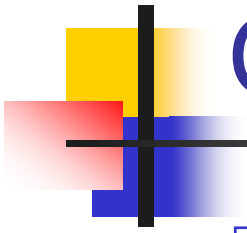


Figure 1.18 Cache Read Operation



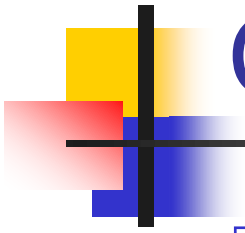
Cache Design

- Cache size
  - Small caches have a significant impact on performance
- Block size
  - The unit of data exchanged between cache and main memory
  - Larger block size more hits until probability of using newly fetched data becomes less than the probability of reusing data that have to be moved out of cache



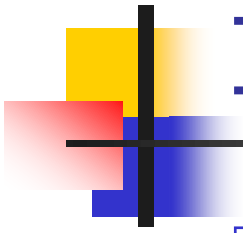
Cache Design

- Mapping function
  - Determines which cache location the block will occupy
- Replacement algorithm
  - Determines which block to replace
  - Least-Recently-Used (LRU) algorithm



Cache Design

- Write policy
  - When the memory write operation takes place
  - Can occur every time block is updated
  - Can occur only when block is replaced
    - Minimizes memory write operations
    - Leaves main memory in an obsolete state

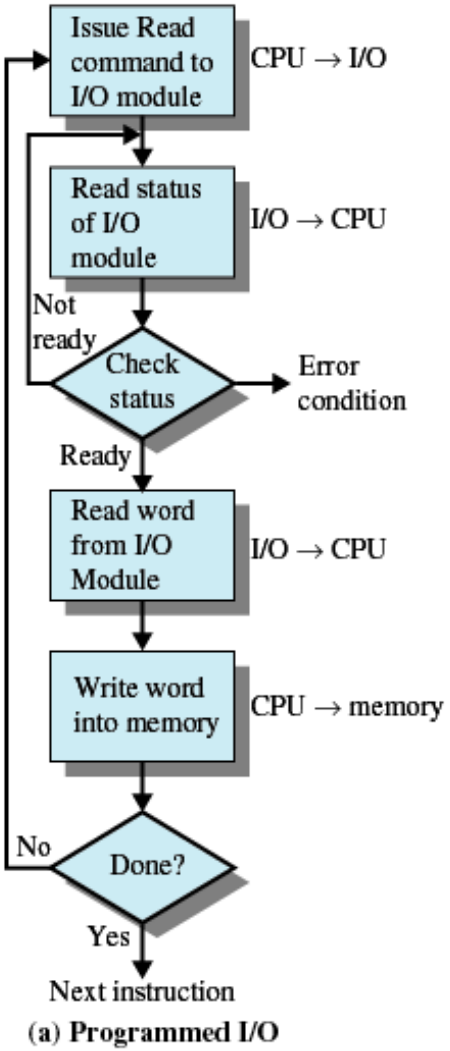


I/O Communication Techniques

- Three techniques are possible for I/O operations:
  - Programmed I/O
  - Interrupt-driven I/O
  - Direct memory access (DMA)

Programmed I/O

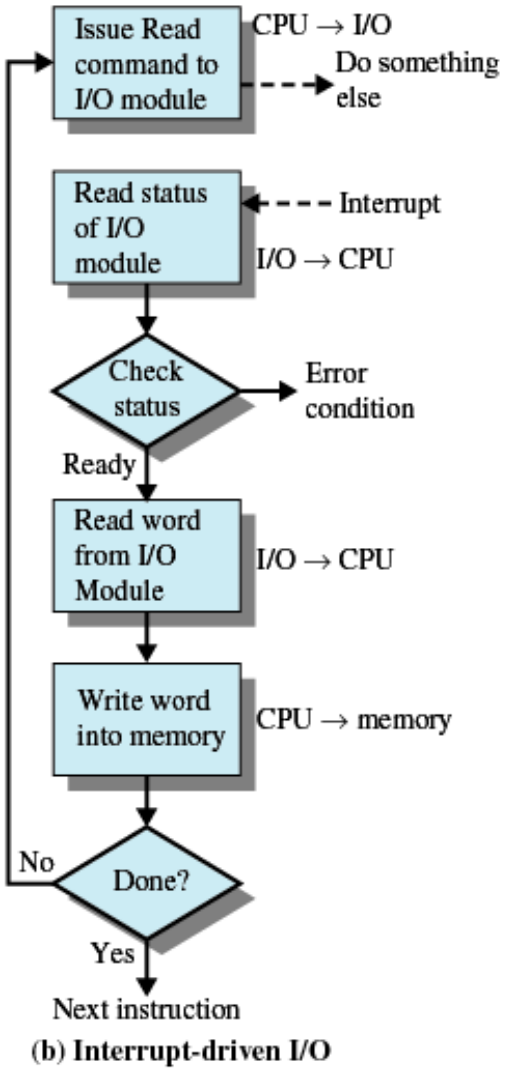
- I/O module performs the action, not the processor
- Sets appropriate bits in the I/O status register
- Processor checks status until operation is complete
- Consumes a lot of processor time because every word read or written passes through the processor



(a) Programmed I/O

Interrupt-Driven I/O

- I/O module issues interrupt request to processor when it is ready to exchange data
- Processor is interrupted
- Processor saves context of program executing and begins executing interrupt-handler
- No needless waiting



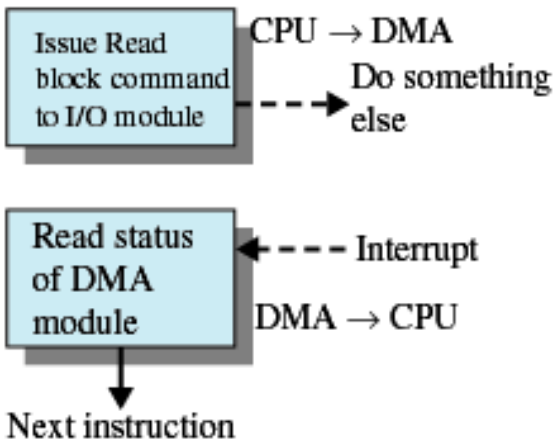
(b) Interrupt-driven I/O

Direct Memory Access (DMA)

- I/O exchanges occur directly with memory
- Processor grants I/O module authority to read from or write to memory
- Relieves the processor responsibility for the exchange

Direct Memory Access

- Transfers a block of data directly to or from memory
- An interrupt is sent when the transfer is complete
- Processor continues with other work



(c) Direct memory access